Everyday Flipper

TVs to Trackers: Basic Tasks for the Flipper Zero



Grey Fox



Who Is This Again?



Grey Fox

- Senior ICS Security Engineer / Wireless Ops Consultant
- Former Air Force (retired)
- Signals Intelligence / Cyberspace Ops / Combat Enabler
- Digital Privacy, RF and Wireless Tech Educator
- Not a lawyer
- Wannabe resistance fighter against Skynet

Do They Have a Plan?



- WHAT is the Flipper Zero?
- WHY is the Flipper Zero?
- Power and buttons
- Menu tour
- Flashing firmware
- Fun with infrared (IR)
- Fun with Bluetooth Low Energy (BLE)
- Scanning for radio signals
- Signal capture and replay
- Touching on Wi-Fi dev boards
- How I use my Flipper Zero

Terms and Breakouts



BLE: Bluetooth Low Energy

GPIO: General Purpose Input Output

IR: Infrared

ISM: Industrial Scientific Medical

NFC: Near Field Communication

RFID: Radio frequency Identification

RX: Reception

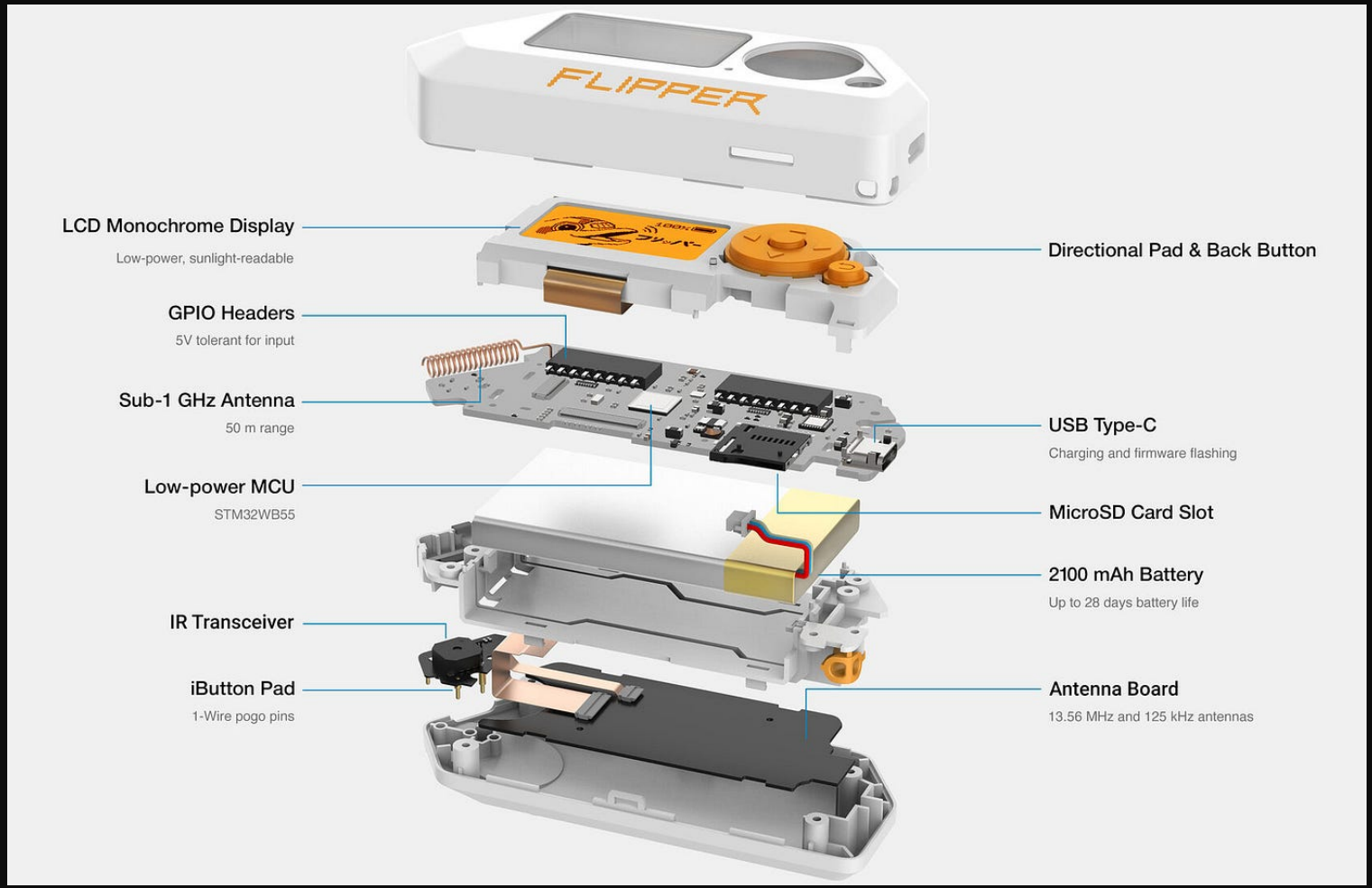
Sub-GHz: Radio frequencies lower than 1000 MHz (1 GHz)

TX: Transmission

I probably forgot a few...

What Is The Flipper Zero?

- All-in-one wireless security testing tool
- Wireless signal library
- Portable tracker, timer, data storage, game device
- BLE, RFID, NFC, IR, USB-C, iButton, GPIO, U2F



Why Is The Flipper Zero?



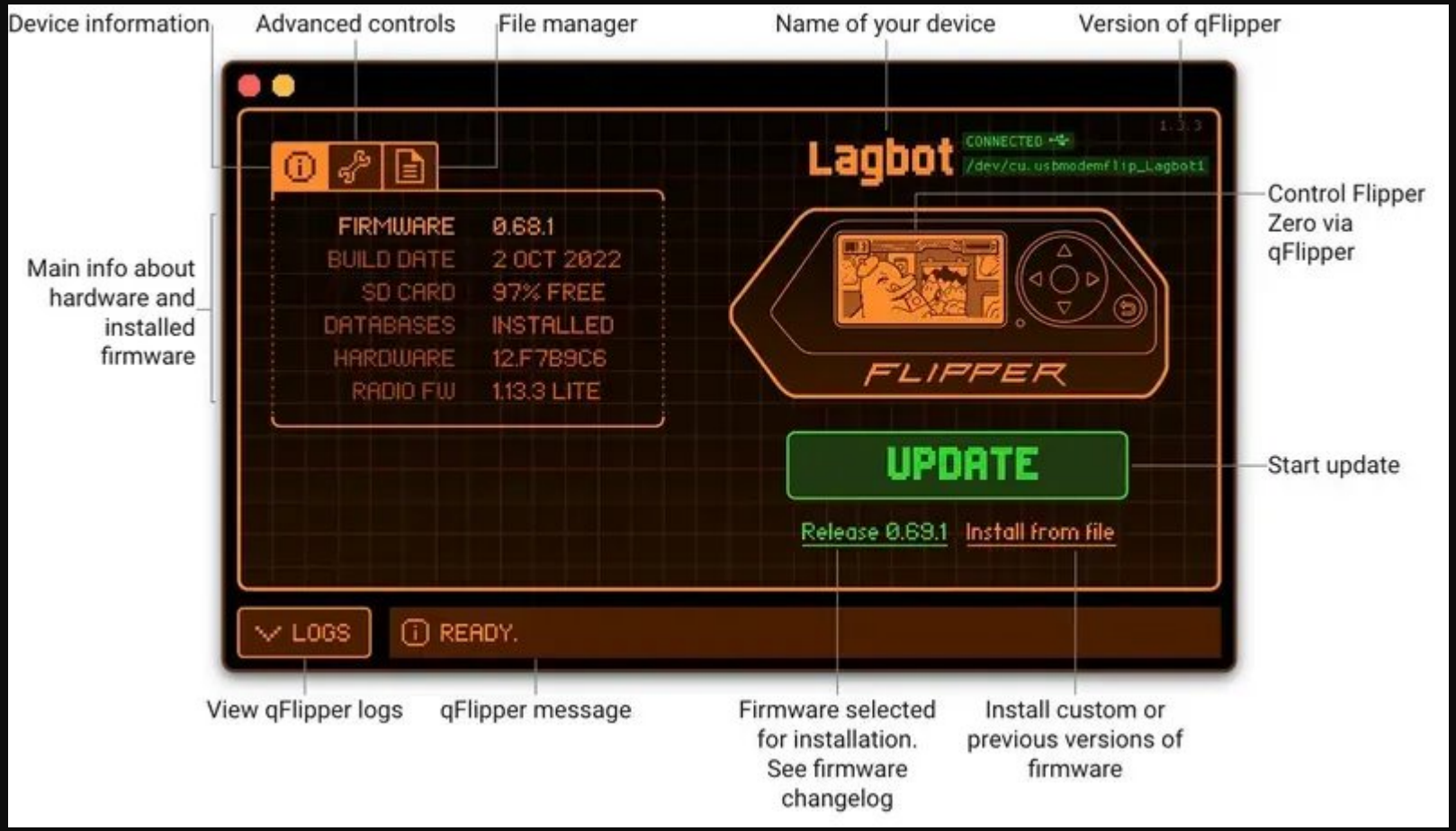
- First of its kind (released summer 2020)
- Open source
- Cheaper than buying everything separately
- Tamagotchi-style animations and gamification
- Lowered the barrier for entry
- Very receptive to community contribution

Let’s Turn It On

- Power on

OPTIONAL

- Install qFlipper
  - Web
  - BadUSB demo



Digging into the Menu

Menu walk-through via qFlipper

Comparing Firmware

- Official firmware (OFW) most supported and “safest”, least features
- Momentum firmware has most customisation features
  - Mostly cosmetic
  - Best security features
- Unleashed has most add-on packages without UI change
- MNTM basically = Unleashed for general use

	OFW	UNLEASHED	MOMENTUM	ROGUE MASTER
Stability of work. `General` [1]	+	+	+	
Regional restrictions are disabled. `Feature` [2]		+	+	+
Easy spoofing for Name, Mac and Serial. `Feature`		+	+	
Rolling Code Support. `Feature`		+	+	+
BLE Spam. `Feature` [3]		+	+	+
Bad USB. `Feature`	+	+	+	+
Bad BT. `Feature`		+	+	+
Subdriving for saving coordinates for Sub-GHz. `Feature` [4]			+	
RGB Backlight. `Decorations` [5]		+	+	+
Advanced Security measures (Lock on Boot, reset on false pins, etc). `Feature` [6]			+	
Interface customization. `Decorations` [7]			+	+
Management App for easy configuration. `Decorations` [8]			+	+
Asset Packs. `Decorations` [9]			+	+

Changing It Up

https://momentum-fw.dev/update

- Full customization
- Better menu
- More NFC capabilities
- More built-in asset packs
- Better device security
- File search

Fun with Infrared



- Brute force remote controls
- Research stored remote controls
- Copy and save IR commands
  - Build remote control clone

Fun with BLE



- Less fun with each device patch
- Requires target device BLE active

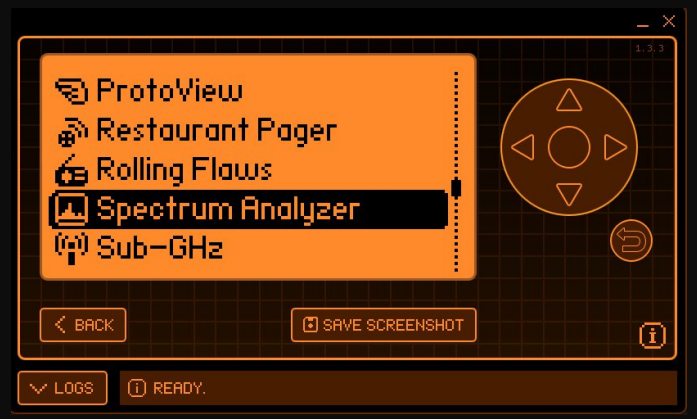
Sub-GHz Basics

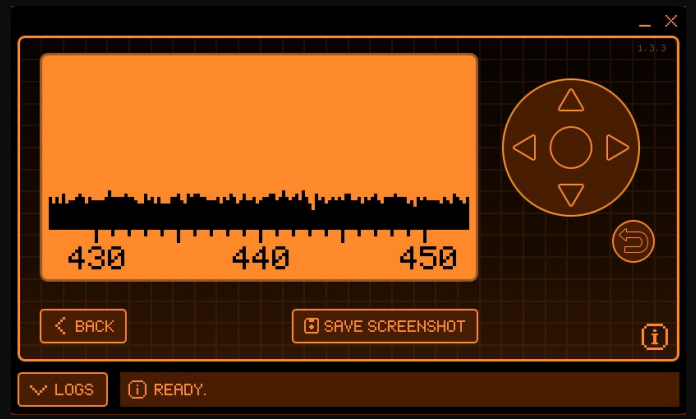
- Sub-GHz signals
  - 315 MHz, 433 MHz, 868 MHz, 915 MHz
- Frequency Analyzer
- Auto-read frequencies

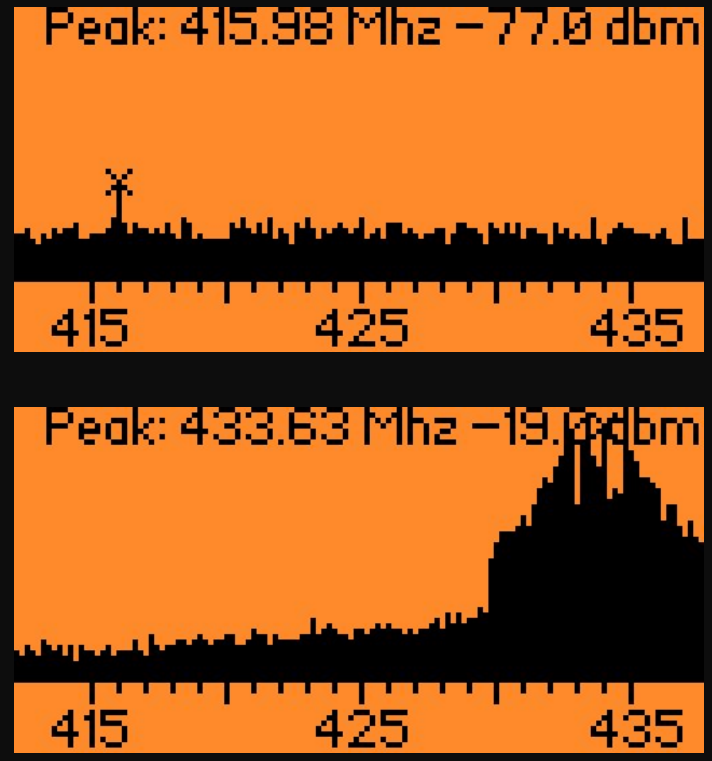


Searching Sub-GHz

- Spectrum Analyzer
- Read raw signals
- Target for Frequency Analyzer or raw signal capture







Raw Signals

- Record and save
- Play back
- Configure for target signal



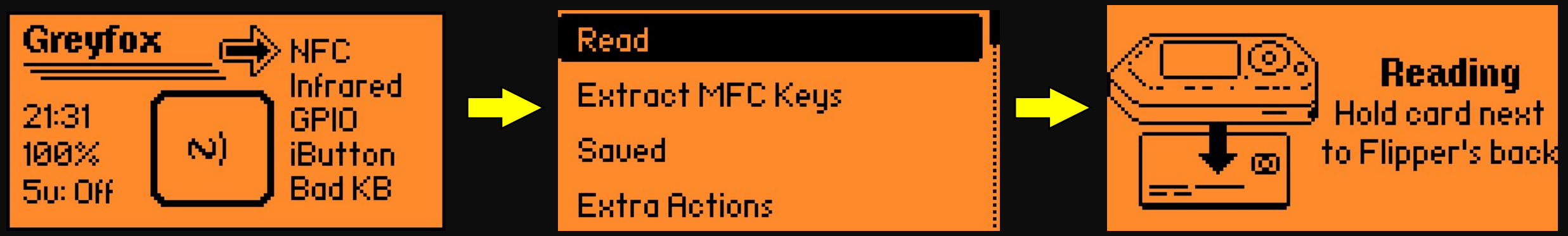
Playing in Sub-GHz



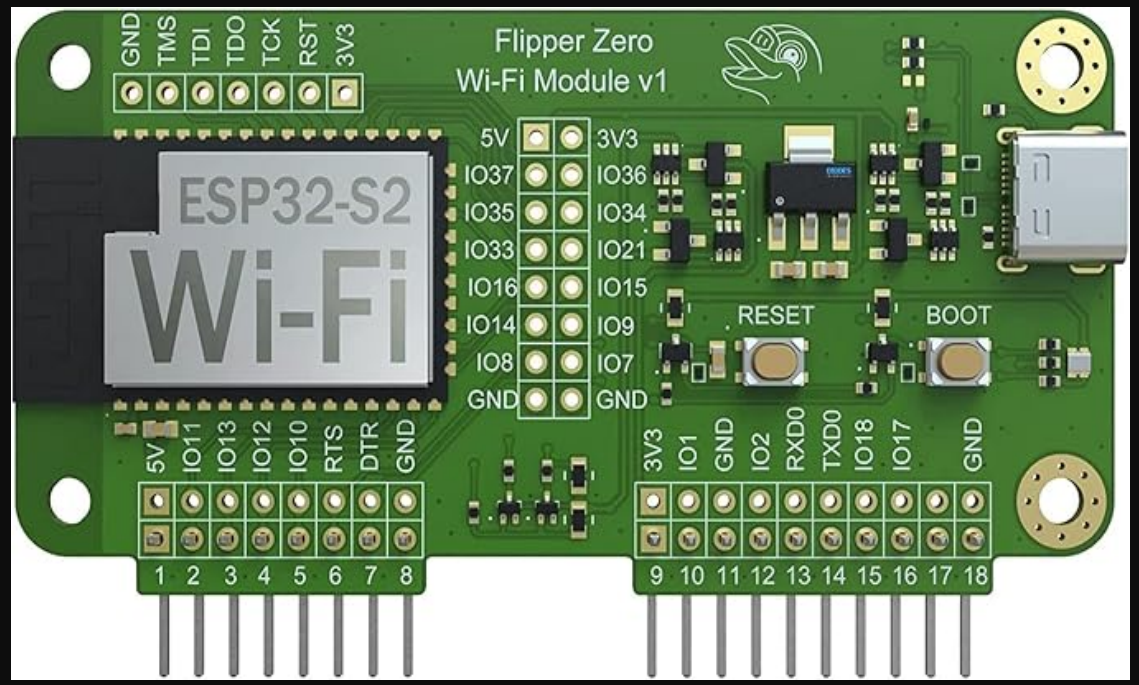
- Zero-security alarms
- Remote control homes
- Vehicle keys
- Weather and temperature sensors
- Handheld radios

Clone Wars (NFC/125 kHz RFID)

- Credit cards
- Hotel room keys (and others)
- Public transportation fare cards
- Resort/Concert bracelets
- Dave & Buster’s

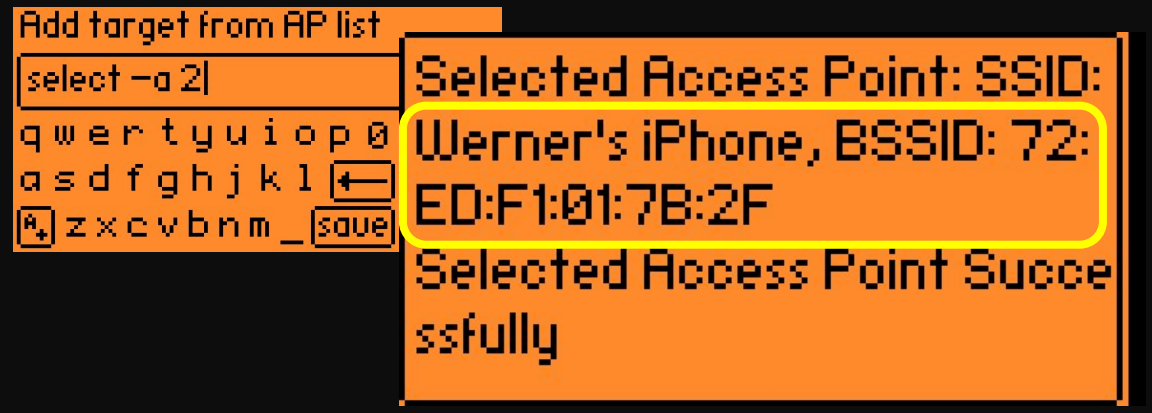


A Brief Moment with Wi-Fi



- Send and receive (1/2 duplex)
- 2.4 GHz
- ~ 20.00 - 60.00 USD
- GPIO / Flipper GUI





Useful Extras

AWOK Dynamics ESP32 v5

Elecrow External CC1101 Antenna for Flipper Zero

SanDisk MAX ENDURANCE microSD™ Card

Thank you